Reliability Monitoring and Predictive Maintenance of Power Electronics with Physics and Data Driven Approach Based on Machine Learning

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***Abstract*—This paper proposes a new prognostics analysis approach for power electronics by combining physics-based and data-driven techniques. Starting with Weibull degradation model, machine learning (ML) techniques are applied to degradation data progressively for continuous reliability monitoring and predictive maintenance decision-making. No prior knowledge of components or mission profiles is required for model training and prediction. Extracted features from analysis can be used to cluster the iteration-based predictions effectively. Another advantage is abrupt change of operation condition can be captured through machine learning for potential lifetime improvement through predictive maintenance. Proposed method can be generalized to other hardware components beyond power electronics.**

***Index Terms*—Power Electronics, Reliability, Predictive Maintenance, Machine Learning**

I. Introduction

As core components in modern energy conversion and control, power electronics devices are being adopted with rising popularity in industrial and consumer products, such as industrial drives, renewable energy systems, electric vehicles (EV), and so on. Research efforts on prognostics and health management (PHM) of power electronics have been proliferating for decades. Effective reliability monitoring and predictive maintenance are critical to ensure equipment reliable operation and personnel safety. A recent survey [1] on Fortune Global 500 manufacturing and industrial firms reports an annual loss of $864 billion from unplanned downtime.

Extensive research has been conducted on power electronics failure mechanism and methods to predict remaining useful lifetime (RUL). [2] provides a comprehensive review on two common failure mechanisms - chip related and package related - and three models for lifetime prediction - empirical model, physics-of-failure model and data-driven model. Recently, machine learning based methods have been widely adopted. [3] summarizes artificial intelligence (AI) application on power electronics with machine learning as a key AI technique and maintenance as a key application area, together with system design and control. Under the machine learning umbrella, various methods can find suitable applications. As an example, Gaussian Process Regression (GPR) is employed in [4] to estimate the remaining lifetime of discrete IGBTs. Existing methods can be grouped into two main categories. One is to generate a closed-form model based on qualification test with substantial amount of units and off-line parameter identification. The generated static model can be loaded in control algorithm thereafter. Units variation can not be captured with this method as it assumes identical lifetime profile for each unit. On the contrary, the other common way is to determine the base structure of prediction model with complete lifetime data from testing limited number of units and the model is dynamically fine-tuned on new units with collected data. This method counts in the unit variation, but still requires prior qualification test. A key challenge in prognostics is to extract useful information from partial data and provide dynamic lifetime prediction. Another challenge, more valuable yet limited attention has been drawn to, is lifetime improvement with predictive maintenance based on RUL prediction.

Previous reliability prediction work provides either physics based [5] or fully data driven [6] methods. In this paper, a novel ensemble approach is proposed by combining physics modeling and data driven methodology. Moreover, analysis with iterative predictions can be used to detect abrupt degradation change and trigger timely inspection and maintenance to potentially improve component lifetime. The remainder of this paper is organized as follows: Section II explains the core algorithm for reliability monitoring with ensemble technique. A real MOSFET aging dataset is used to validate the algorithm and features are discussed for operation condition monitoring in Section III. Section IV demonstrates results clustering and data compression with machine learning. Finally, Section V summarizes proposed work and directions for future research.

II. Algorithm Description

A. Weibull Distribution for Reliability Engineering

Weibull distribution has been one of the most widely utilized techniques in reliability engineering due to its flexibility to model the wide family of degradation profiles and superior interpretability of distribution parameters [7]. For reference, Eqns. 1.1 – 1.3 lists key equations related to Weibull distribution. Shape parameter β determines the component failure rate change with time and $\beta > 1$ is the most common case, which indicates failure rate increases with time and is the most general degradation pattern. $F(t)$ implies the accumulated degradation and survival function $R(t)$ can be interpreted as RUL indicator, which is adopted as the base for reliability modeling and analysis in this paper.

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$$f(t) = \frac{\beta}{\alpha}(\frac{t}{\alpha})^{\beta-1}e^{-(t/\alpha)^{\beta}} \tag{1.1}$$

$$F(t) = 1 - e^{-(t/\alpha)^{\beta}} \tag{1.2}$$

$$R(t) = 1 - F(t) = e^{-(t/\alpha)^{\beta}} \tag{1.3}$$

where:

α: scale parameter
β: shape parameter
$f(t)$: probability density function (PDF)
$F(t)$: cumulative distribution function (CDF)
$R(t)$: survival function (SF)

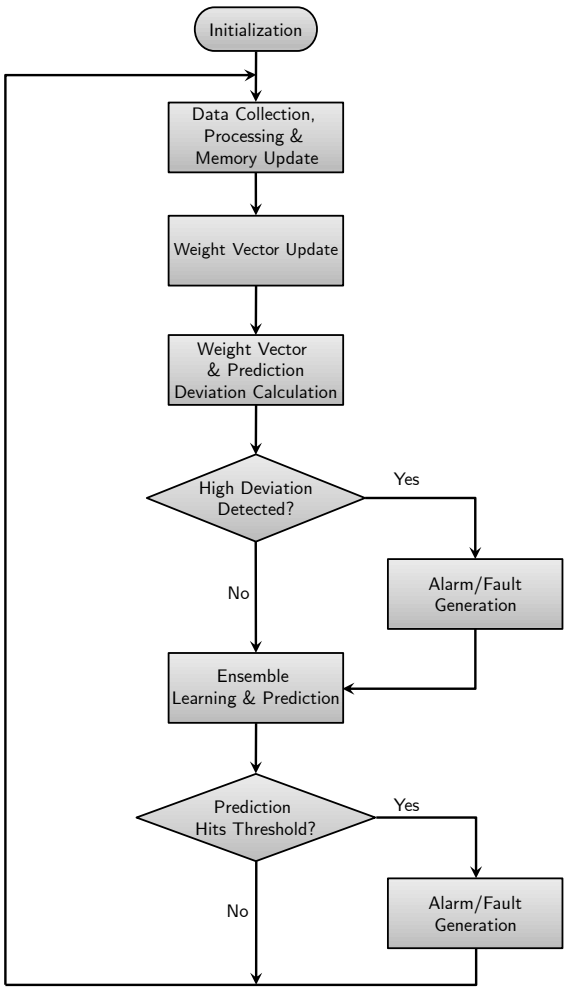

Fig. 1. Algorithm flowchart

B. Algorithm for Reliability Monitoring

The flowchart in Fig. 1 demonstrates the algorithm for reliability analysis and prediction, with underlying equations listed in Eqns. 2 – 4. For a component, an observable variable needs to be identified as the indicator for reliability, for example, MOSFET conduction state resistance Rds_{ON} or IGBT conduction state voltage drop Vce_{ON}. Selected indicator is used as input to the algorithm. Multiple independent predictors are trained with collected data of different window sizes and provide reliability predictions progressively. Training is performed through non-linear regression on survival function to identify the best parameters for each predictor. In addition to point prediction, uncertainty is calculated to generate a confidence interval (CI) based on required confidence level. Point prediction and uncertainty from individual predictors are combined to generate an ensemble prediction with uncertainty via a weight vector. Deviation calculation is performed on both prediction accuracy and weight vector. For prediction accuracy, prediction results of each predictor are compared against the newly collected real data and root mean square error (RMSE) is calculated as metric. Weight vector is dynamically updated based on prediction performance of each predictor where

$$CI(t)^{(j)} = R(t)^{(j)} \pm \sigma_R(t)^{(j)} \tag{2.1}$$

$$R(t)^{(j)} = \sum_{i=1}^{L} \theta_i^{(j)} R(t)_i^{(j)} \tag{2.2}$$

$$\sigma_R(t)^{(j)} = \sum_{i=1}^{L} \theta_i^{(j)} \sigma_R(t)_i^{(j)} \tag{2.3}$$

$$\boldsymbol{\theta}^{(j)} = [\theta_1^{(j)}, \theta_2^{(j)}, ..., \theta_L^{(j)}]^T \tag{3.1}$$

$$\theta_i^{(j)} = \frac{1}{L-1}\left(1 - \frac{\text{RMSE}_i^{(j)}}{\sum_{k=1}^{L}\text{RMSE}_k^{(j)}}\right) \tag{3.2}$$

$$\text{RMSE}_i^{(j)} = \sqrt{\frac{\sum_{k=1}^{N}(y_k^{(j)} - R(t)_{ik}^{(j)})^2}{N}} \tag{3.3}$$

$$\text{RMSE}_{\Sigma}^{(j)} = \sum_{i=1}^{L}\text{RMSE}_i^{(j)} \tag{4.1}$$

$$\Delta R_{mn}^{(j)} = \frac{\sum_{k=1}^{N}|R(t)_{mk}^{(j)} - R(t)_{nk}^{(j)}|}{N} \tag{4.2}$$

where:

L: number of total predictors
$CI(t)^{(j)}$: combined prediction confidence interval on iteration j
$R(t)^{(j)}$: combined point prediction on iteration j
$\sigma_R(t)^{(j)}$: combined uncertainty band on iteration j
$R(t)_i^{(j)}$: point prediction by predictor $i$on iteration j
$\sigma_R(t)_i^{(j)}$: uncertainty band by predictor i on iteration j
$\boldsymbol{\theta}^{(j)}$: weight vector on iteration j
$\theta_i^{(j)}$: weight parameter for predictor i on iteration j
$\text{RMSE}_{\Sigma}^{(j)}$: sum of RMSE of each predictor on iteration j
$\text{RMSE}_i^{(j)}$: root mean square error for predictor i on iteration j
N: prediction window size
$y_k^{(j)}$: measured k-th new data point on iteration j
$R(t)_{ik}^{(j)}$: predicted k-th data point by predictor i on iteration j
$\Delta R_{mn}^{(j)}$: mean absolute error of predictions by predictors m and n on iteration j

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higher accuracy leads to increased weight for next iteration and vice versa. When abnormal deviation beyond a preset limit is detected, alarm/fault can be triggered for operation condition check and evaluation.

Ensemble reliability prediction is summarized in Eqns. 2.1 – 2.3. Overall confidence interval $CI(t)^{(j)}$ of predicted reliability on iteration j consists of combined point prediction $R(t)^{(j)}$ and combined uncertainty prediction $\sigma_R(t)^{(j)}$. Dynamic update of weight vector $\boldsymbol{\theta}^{(j)}$ based on prediction performance is the core of this learning algorithm. RMSE between predicted results and collected new data, as in Eqn. 3.3, is adopted as the metric for weight update. Individual weight parameter $\theta_i^{(j)}$ is updated with the proposed Eqn. 3.2 to ensure $\Sigma\boldsymbol{\theta}^{(j)} = 1$. It is a penalty based update scheme where the relative prediction performance determines the weight of each predictor for the next iteration. In addition to the separate deviation between individual prediction and real data denoted by $\text{RMSE}_i^{(j)}$, two overall metrics are calculated in Eqns. 4.1 and 4.2 for degradation profile tracking and predictive maintenance alert generation – $\text{RMSE}_\Sigma^{(j)}$ for all predictors and relative deviation $\Delta R_{mn}^{(j)}$ for all predictor pairs, where $m, n \in [1, 2, ..., L],\ m \neq n$.

To validate the algorithm, Fig. 2 demonstrates the implementation on one iteration with a simulated degradation curve and sampled data points with artificial Gaussian noise. Three set of α's and β's are used to create the concatenated profile to simulate the everchanging degradation process, with concatenation points at Time 23 and 38. 3 predictors are selected with training data of different window sizes. Predictor 1 uses the most recent 45 points, predictor 2 uses the most recent 30 points and predictor 3 with the most recent 15 points. Non-linear least-square regression is practiced on each predictor with the corresponding windowed data to generate the optimal prediction model as illustrated by the solid lines. Individual reliability prediction from each predictor, illustrated by the dashed line, for a future time window (10 points here) is calculated along with uncertainty of a pre-determined confidence level, 95% (2σ) here. Combined prediction and uncertainty is eventually generated by combining individual

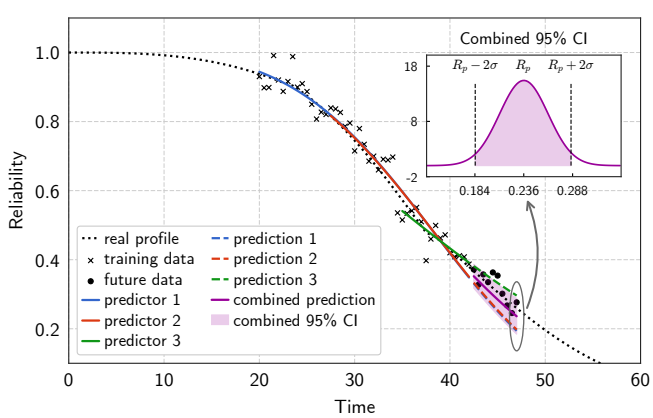

Fig. 2. Algorithm demonstration on one iteration

predictions and uncertainties with updated weight vector $\boldsymbol{\theta}^{(j)}$. The inset demonstrates the distribution of predicted reliability, with a RUL of 23.6% and the value will stay in [0.184, 0.288] with a confidence level of 95%. Number of predictors, training data window sizes and confidence level can be selected based on application requirements, allocated computation resources, etc.

Another prominent feature of this method is the flexibility to maintain tracking stability while monitoring abrupt change in degradation profile. In Fig. 2, predictor 1 with the longest training data window maintains the tracking stability, and predictor 3 provides detection capability for abrupt profile variation. As shown here, predictor 3 is able to capture the sudden change starting at Time 38 due to minimal influence from shortest window of history data and improve the overall prediction towards real profile. The superior prediction performance from predictor 3 on this iteration will promote its weight θ_3 used in next iteration. Ensemble approach through evaluation and update of weight vector on each iteration guarantees timely tracking and prediction of reliability and multiple predictors improve prediction accuracy. This "training–predicting–weight updating" procedure will be executed progressively through the component lifetime.

III. Validation with MOSFET Aging Data

Power MOSFET has been among the most widely used power electronics devices for decades. Extensive research work has been conducted regarding the RUL prediction and device-level protection to improve lifetime. [8] proposes an exponential degradation model to predict MOSFET RUL based on least-square curve fitting with ON state resistance Rds_{ON} as degradation precursor. Various protection schemes were proposed to reduce the damage mainly from short-circuit incurred overcurrent and prolong device lifetime, for example, the current mirror based overcurrent protection [9] and desaturation detection based overcurrent protection [10].

To validate the proposed algorithm, an experimental MOSFET aging dataset shared by NASA Prognostics Center of Excellence is used in this paper. Experimental setup and detailed information about the data collection can be found in [11]. ON state resistance Rds_{ON} is extracted from dataset and processed as reliability indicator with Eqn. 5, where $Rds_{ON}(0)$ is the resistance at the beginning of test.

$$R(t) = \frac{2Rds_{ON}(0) - Rds_{ON}(t)}{Rds_{ON}(0)} \tag{5}$$

First subplot in Fig. 3 is the processed reliability curve down to 0.92. The rest demonstrate the "training–predicting–weight updating" procedure for 14 iterations. In this validation process, 3 predictors are used for ensemble prediction with training window sizes selected as 450, 300 and 150 points. Prediction window size is set to 150 points. On a iteration-to-iteration basis, the algorithm provides self-adjusted tracking and prediction of degradation profile via dynamic learning through collected data and update of weight vector. Overall, proposed ensemble prediction provides good prediction of the

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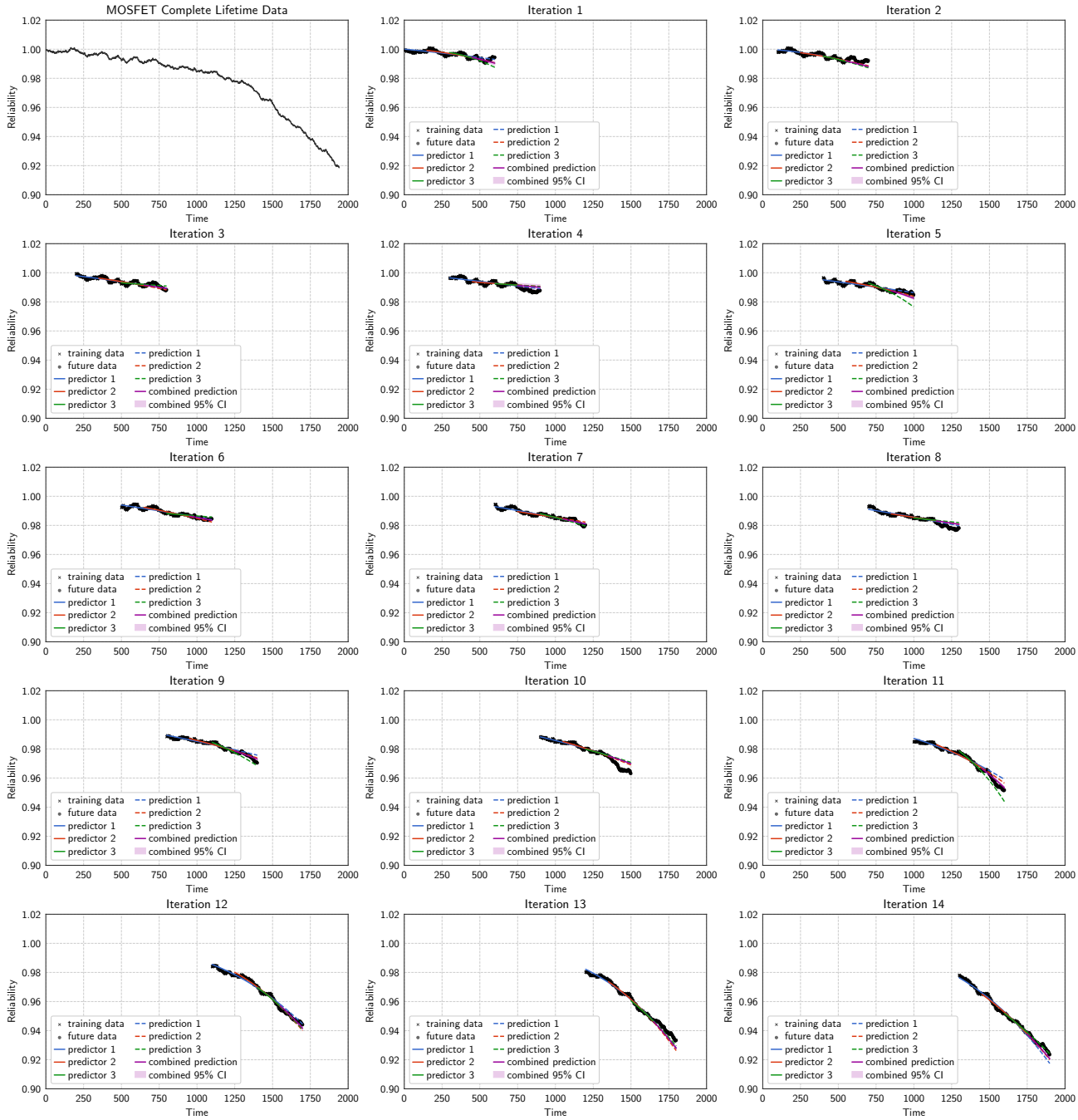

Fig. 3. Progressive analysis and prediction with MOSFET aging data

real degradation curve. By inspecting the calculated metrics, weight vector and total RMSE, some abnormal iterations can be determined easily and confirmed by corresponding iteration subplots in Fig. 3.

From weight vector update Fig. 4 it can be observed that in iteration 5, updated weight θ_3 is considerably lower than θ_1 and θ_2. It is the result of lower prediction accuracy from predictor 3 compared with that from predictor 1 and 2. Corresponding subplot in Fig. 3 demonstrates exactly the described pattern. Knowledge regarding the prediction performance can also be obtained by checking the calculated total RMSE in Fig. 5. Iteration 10 stands out with the highest total RMSE, which implies high prediction error from at least one predictor. Corresponding subplot in Fig. 3 shows all 3 predictions are off the real curve, although 3 predictions stay very close to each other.

Relative deviations among individual predictions calculated as mean absolute error (MAE) of prediction pairs in Eqn. 4.2

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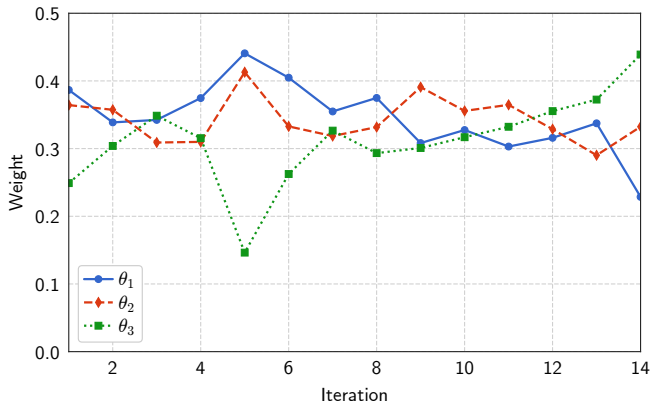

Fig. 4. Weight update on each iteration

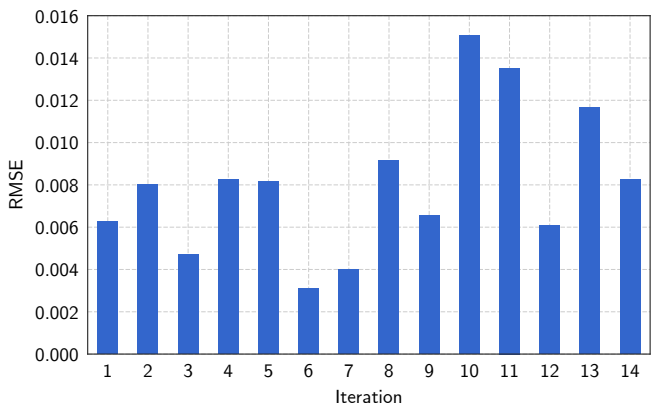

Fig. 5. Total RMSE on each iteration

indicate predictions discrepancy and imply potential abrupt change in operation environment (temperature, humidity, etc.). High deviations should trigger alerts for operation condition check and maintenance evaluation. On the other hand, visually checking the weight and RMSE deviation is inefficient and can be deceptive in certain cases. Iteration 11, for instance, is a favorable scenario with combined prediction falls well in the individual predictions. However, a high total RMSE is still read from Fig. 5 and can cause a false alarm. In addition to per-iteration analysis, grouping iterations with similar pattern can facilitate revealing common underlying characteristics which is difficult to uncover from individual iteration. Therefore, intelligence is needed to support "smarter" automated anomaly detection and pattern grouping, which is the topic of next section.

IV. Results Clustering and Decision-making with Machine Learning

To automatically group the results of MOSFET validation from section III, non-supervised K-means algorithm is adopted with cluster hyperparameter set to 4. Absolute and relative deviations are selected as input features for clustering, 6 features in total for 3 predictors. This machine learning technique generates surprisingly good clustering results demonstrated by the fact that iterations within each group exhibit very similar pattern while different groups show distinct patterns. Take group 1 as example, for each iteration all 3 predictions deviate from the real profile on the same side and none of the predictors captures the degradation profile. Potential causes include bad model training, unfavorable selection of parameters such as the window sizes and number of predictors, or actual sudden change of degradation profile. Complete clustering results, patterns and derived decisions for action are summarized in Table I. Four scenarios are created based on prediction performance: abnormal, acceptable, normal and favorable. After the iterations are properly clustered, knowledge from weight update as in Fig. 4 and total RMSE as in Fig. 5 can be incorporated to determine the extent of deviation for improved decision-making.

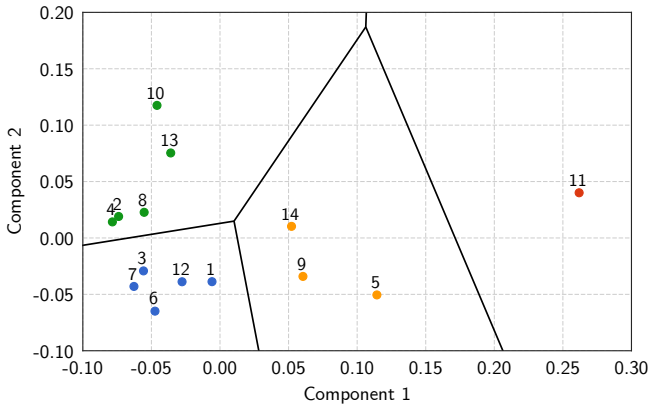

Fig. 6. Clustering with PCA results

From the perspective of algorithm implementation, data storage and transfer is a common challenge in industrial applications especially when running centralized analysis for multiple distributed components and high data traffic travels through the network. Effectively compressing the data while reserving valuable information to the most extent is a natural choice. Another non-supervised machine learning technique, principal component analysis (PCA) is adopted for data transformation on the selected 6-dimension (6D) features – absolute and relative deviations – into 2D space. Fig. 6 shows clustering results with the compressed 2D data. Iteration number is annotated next to each point. The color code is based on clustering results with original 6D features as in Table I. With a 66.7% compression ratio, identical clustering results are generated as those with original data, which implies majority of the useful information is reserved during data compression. Cluster boundaries are also plotted on the new 2D plane and this is another useful PCA characteristic for this application – visualizing features in a lower-dimension space.

V. Conclusions and Future Work

An innovative method for reliability monitoring and predictive maintenance is presented by combining degradation physics and data-driven techniques. Ensemble reliability prediction is achieved through multiple predictors and ensures

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TABLE I. Iterations clustering and derived decisions

Cluster	Iterations	Pattern	Decision
1	2, 4, 8, 10, 13	all predictions deviated from observed data in one direction	abnormal
2	5, 9, 14	one prediction deviated from observed data; two stay close to observed data	acceptable
3	11	all predictions deviated from observed data in both directions with high RMSE	normal
4	1, 3, 6, 7, 12	all predictions stay close to observed data	favorable

tracking stability and capability to capture abrupt degradation changes. In addition to point prediction, uncertainty is also provided based on confidence level for better decision-making. As core of the algorithm, weight vector is updated via dynamic learning from new data and prediction performance evaluation. Multiple metrics are calculated based on RMSE for degradation profile monitoring and anomaly detection. Algorithm is validated with real MOSFET aging data and exemplary methodology to convert measurable variable to reliability indicator is illustrated. It demonstrates close tracking and prediction of degradation profile. Machine learning techniques are practiced for pattern identification, decision-making and data compression. K-means is competent to cluster iterations with similar pattern. Patterns are summarized for all groups and corresponding decisions are derived. Proactive maintenance can be performed based on the decision to potentially improve component lifetime. With the advantages of dynamic learning based on partial data and degradation mechanism independent nature, this method can be applied to other hardware components as long as the degradation process follows Weibull pattern. Future research areas include data processing, outliers removal, and algorithm generalization to components beyond power electronics.

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